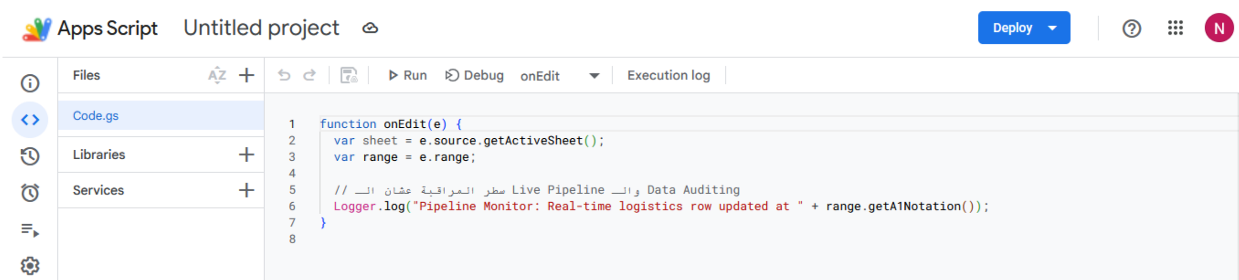
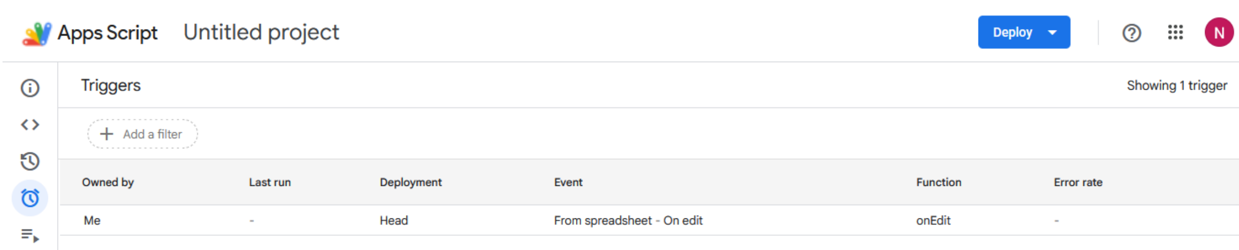
Data Collection & Automation

Automation Experiment

To explore automated data collection workflows, the dataset was uploaded to Google Sheets. Google Apps Script was used to automate data handling tasks, and time-driven triggers were configured to execute the script automatically.

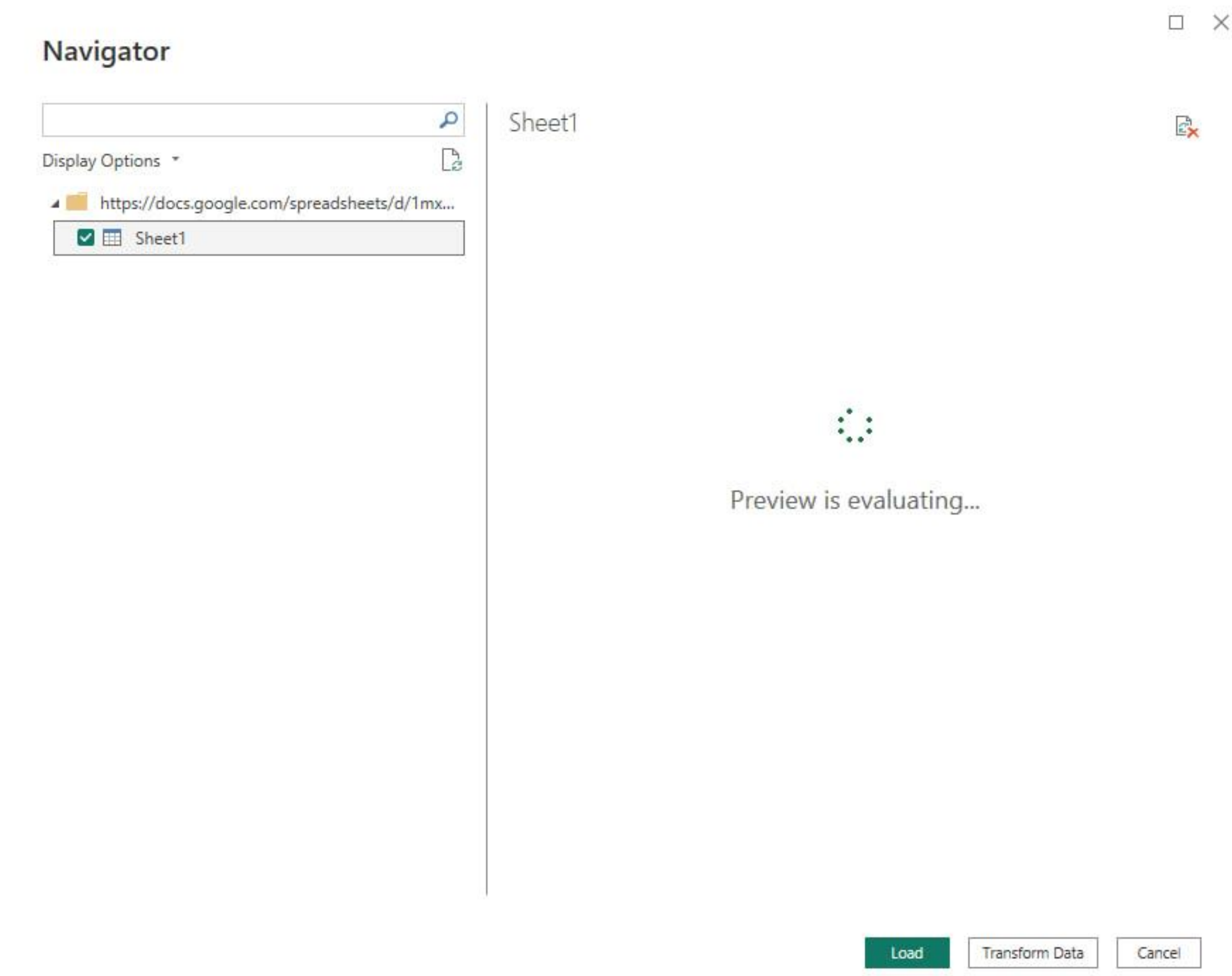
The automation layer was implemented as a proof of concept to demonstrate integration capabilities between cloud-based data sources and business intelligence tools.





Integration Testing

Power BI integration with Google Sheets was tested successfully.



However, due to the large dataset size (approximately 180,000 records and 53 columns), refresh and loading performance became impractical for analytical workloads. Data refresh operations required excessive loading time and negatively impacted the overall analytical experience.

Production Analytics Environment

To ensure efficient data processing and reporting performance, the final analytical solution was developed using the original dataset source locally.

Google Sheets remained part of the automation and integration validation phase, while Power BI was used for data preparation, modeling, DAX calculations, and dashboard development.

Data Understanding & Preparation

Dataset Overview

The dataset was initially provided as a single denormalized table containing information related to customers, products, categories, departments, orders, and shipping activities.

A detailed exploration phase was conducted to understand the dataset structure, identify redundancies, validate business entities, and prepare the data for analytical modeling.

Data Quality Assessment

Several columns were examined to identify duplicate attributes and redundant information.

A row-level comparison was performed across the entire dataset, revealing the following duplicated fields:

- Customer ID and Order Customer ID
- Product Card ID and Order Item Card Product ID
- Category ID and Product Category ID
- Benefit Per Order and Order Profit Per Order

After validation, duplicate columns were removed to simplify the model and eliminate unnecessary redundancy.

Data Cleaning

Several attributes were identified as non-essential for analytical reporting and business decision-making.

The following fields were removed from the analytical model:

- Customer Email
- Customer Password
- Customer First Name
- Customer Last Name
- Customer Street
- Product Image
- Product Description

Removing these attributes reduced model complexity and improved overall efficiency while preserving all business-relevant information.

Business Terminology Standardization

To improve readability and make business metrics easier to understand, several columns were renamed.

Original Name	New Name
Sales	Gross Sales
Order Item Total	Net Sales
Order Profit Per Order	Item Profit

The updated terminology better reflects the actual business meaning of each measure and improves communication with business stakeholders.

Business Process Analysis

The transactional grain of the dataset was carefully analyzed.

The analysis showed that a single order may contain multiple products, resulting in multiple records sharing the same Order ID while representing different purchased items.

Based on this observation, the dataset was logically separated into two distinct business processes:

- Orders
- Order Items

This separation reduced unnecessary repetition of order-level information and laid the foundation for a scalable analytical model.

Data Modeling & Relationship Design

Data Normalization

Rather than relying on the original flat structure, a complete analytical model was designed from scratch based on business entities identified during the exploration phase.

The original dataset was transformed into the following logical entities:

- OrderItems
- DimOrders
- DimCustomers
- DimProducts
- DimCategory

- DimDepartment
- Date

Each entity was manually designed after analyzing business requirements, validating relationships, and determining the correct granularity level.

Duplicate records were removed from dimension tables using their corresponding business keys.

Data types were standardized across the model, and all identifier fields and postal codes were converted to text data types to prevent incorrect aggregation and preserve their categorical nature.

Data Validation

Before implementing relationships, a comprehensive validation process was performed.

The following checks were completed:

- Missing value analysis
- Consistency validation
- Range validation
- Attribute verification

The analysis showed that most fields contained complete and reliable information. The only attribute with a significant number of missing values was Order Zip Code.

Consistency validation confirmed that order-level attributes remained stable within each order, including shipping information, customer assignments, and order characteristics.

Range analysis revealed that the dataset covers transactions between:

- 06 January 2015
- 31 January 2018

This period defines the analytical coverage of the model.

Feature Engineering

A new calculated attribute named **DelayDays** was created in Power Query.

The measure calculates the difference between actual shipping duration and scheduled shipping duration, enabling shipment performance evaluation and delay analysis.

Product Hierarchy Assessment

Product-related attributes were reviewed to validate product hierarchy consistency.

The Product Status field contained only a single value (0), indicating that all products were active throughout the analysis period.

The final product hierarchy consisted of:

- 118 Products
- 51 Categories
- 11 Departments

Date Dimension

A dedicated Date dimension was created using the CALENDAR function.

The calendar range was dynamically generated using the minimum order date and maximum shipping date available in the dataset.

Additional temporal attributes were created, including:

- Year
- Quarter
- Month

The table was designated as the official Date Table to support accurate time-intelligence calculations and reporting.

Snowflake Schema Design

The analytical model was intentionally designed using a Snowflake Schema approach.

Instead of connecting all dimensions directly to a central fact table, hierarchical entities were normalized into multiple related dimensions.

This approach reduced redundancy and preserved the natural business hierarchy between departments, categories, products, orders, and customers.

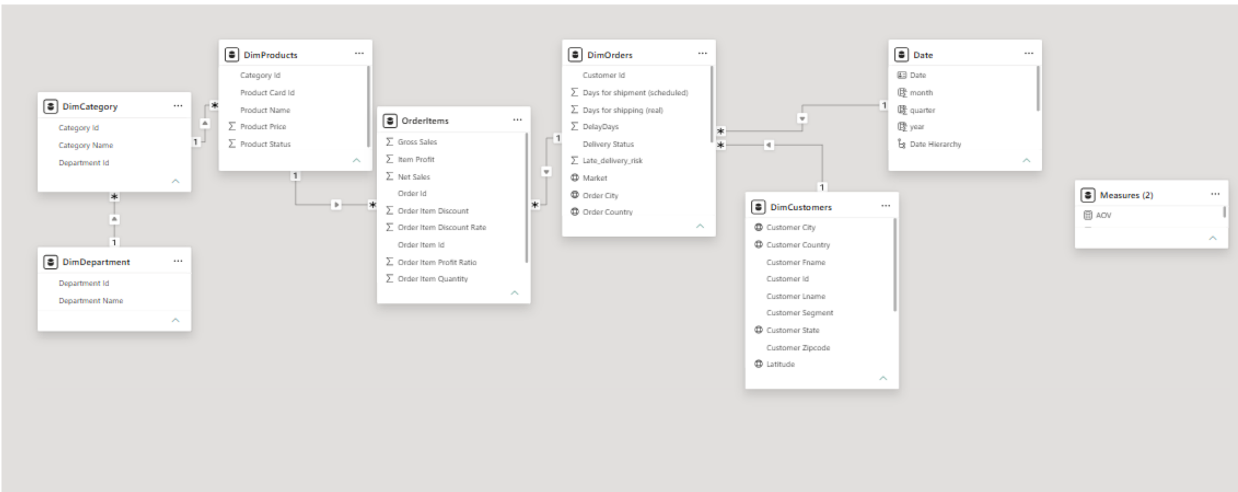
The final relationship structure was implemented as follows:

- Date → DimOrders
- DimCustomers → DimOrders
- DimOrders → OrderItems

- DimProducts → OrderItems
- DimCategory → DimProducts
- DimDepartment → DimCategory

All relationships were configured as one-to-many relationships using validated business keys.

The entire schema was designed and implemented manually based on business analysis and entity relationships rather than directly adopting the original source structure.



Geographic Configuration

Geographic attributes were configured using Power BI Data Categories.

Relevant location fields were assigned appropriate geographic classifications to improve map visualizations and support geographic analysis throughout the dashboard.

Model Optimization

After completing the transformation and modeling phases, the original source table was excluded from the final model using the Disable Load option.

This optimization reduced memory consumption, improved model performance, and maintained the source query for future validation and maintenance activities.

Dashboard Development & Business Insights

Dashboard Objectives

The dashboard was designed to support business and operational decision-making across the supply chain process.

The reporting solution was divided into two primary analytical perspectives:

Business Performance Overview

Focused on revenue generation, profitability, customer contribution, product performance, and market performance.

Logistics Performance & Risk Analysis

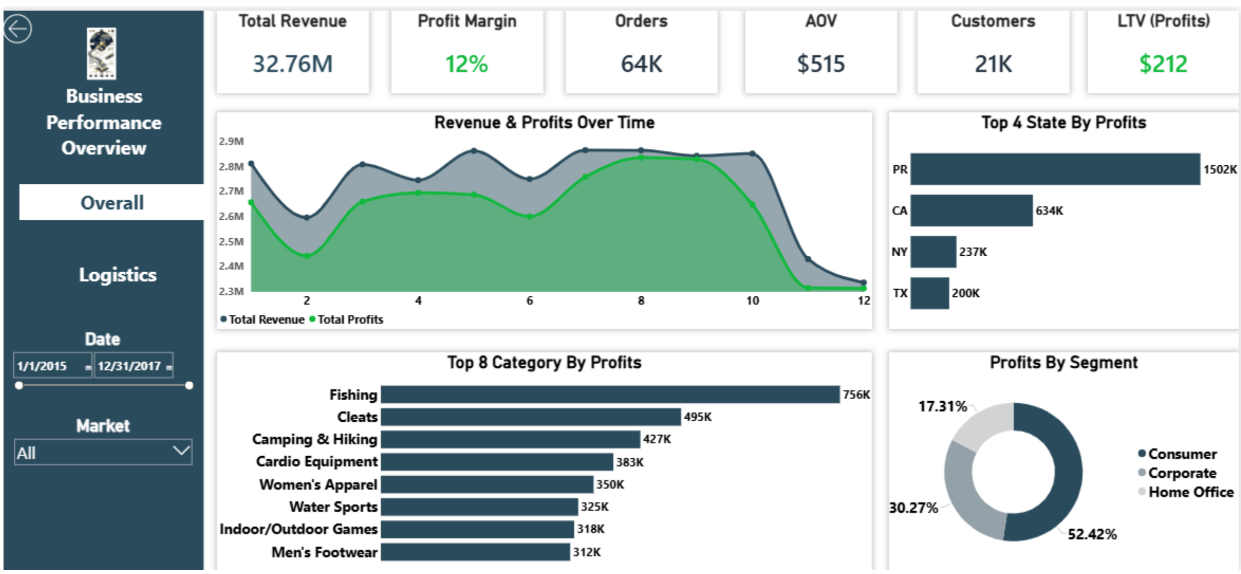
Focused on operational risks, delivery performance, fraud detection, order fulfillment issues, and loss-making products.

This separation allows business users to evaluate both financial performance and operational efficiency independently while maintaining a unified analytical experience.

Business Performance Overview

The first dashboard page was designed to answer a fundamental business question:

Where is the company generating revenue and profit?



Key Performance Indicators (KPIs)

KPI	Value
Total Revenue	$32.76M
Profit Margin	12%

KPI	Value
Orders	64K
Average Order Value (AOV)	$515
Customers	21K
Customer Lifetime Value (LTV)	$212

These metrics provide an executive-level overview of overall business performance.

Revenue & Profit Trend Analysis

Revenue and profit trends were analyzed across the available time period.

The analysis enables stakeholders to monitor business growth patterns and identify periods of stronger or weaker performance.

Geographic Profitability Analysis

Profit contribution was evaluated across different states.

The four most profitable states were:

- Puerto Rico
- California
- New York
- Texas

A significant profitability gap was observed after the fourth-ranked state, therefore the visualization focused on the most impactful contributors.

Category Performance Analysis

The leading profit-generating categories included:

- Fishing
- Cleats
- Camping & Hiking
- Cardio Equipment
- Women's Apparel

This analysis supports merchandising and inventory planning decisions.

Customer Segment Analysis

Customer segments were evaluated according to their contribution to total profit.

The analysis revealed that:

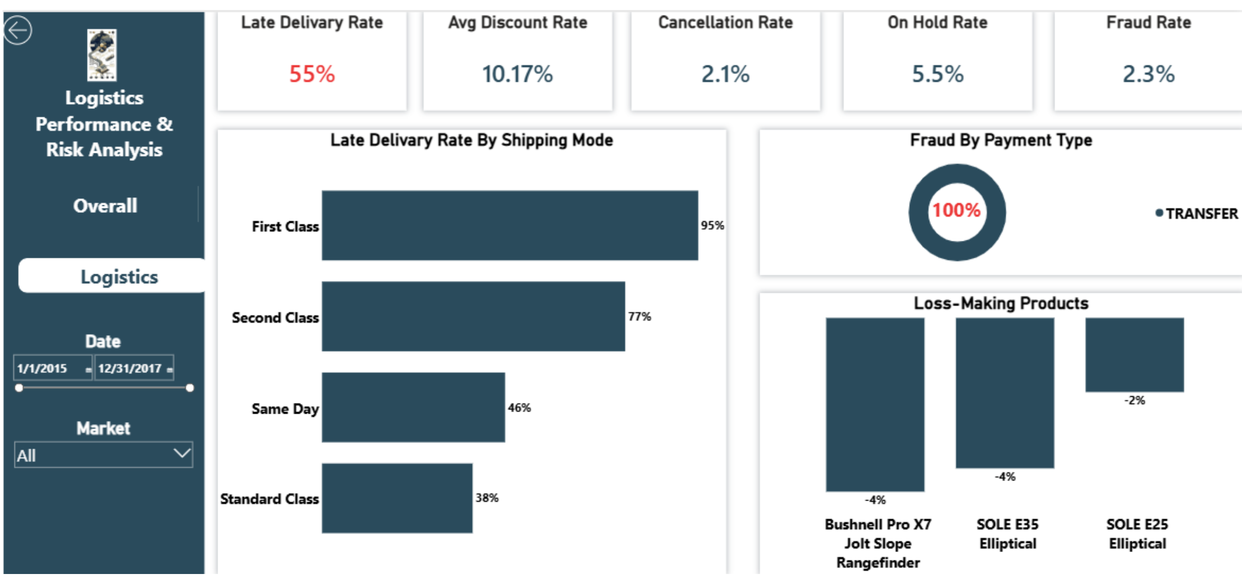
- Consumer customers generated approximately 52% of total profits.
- Corporate customers generated approximately 30% of total profits.
- Home Office customers generated approximately 17% of total profits.

These insights help identify the most valuable customer groups and support customer acquisition strategies.

Logistics Performance & Risk Analysis

The second dashboard page was designed to identify operational inefficiencies, business risks, and supply chain performance issues.

The objective was to move beyond descriptive reporting and perform root-cause-oriented analysis.



Operational Risk KPIs

KPI	Value
Late Delivery Rate	55%

KPI	Value
Average Discount Rate	10.17%
Cancellation Rate	2.1%
On Hold Rate	5.5%
Fraud Rate	2.3%

These metrics provide visibility into operational performance and potential business risks.

Delivery Performance Analysis

The analysis revealed that:

55% of all orders were delivered late.

This represents the most significant operational issue identified during the project.

To investigate the root cause, delivery performance was analyzed across shipping methods.

Shipping Mode	Late Delivery Rate
First Class	95%
Second Class	77%
Same Day	46%
Standard Class	38%

Despite being positioned as the fastest shipping option, First Class exhibited the highest delay rate.

This finding indicates a potential operational bottleneck within premium shipping services and warrants further business investigation.

Fraud Analysis

Fraud-related orders represented approximately 2.3% of all transactions.

Payment method analysis revealed that:

100% of identified fraudulent orders were associated with bank transfer payments.

This finding suggests that bank transfer transactions may require additional monitoring, verification controls, or fraud-prevention measures.

Loss-Making Product Analysis

The analysis identified three products operating at a loss:

- Bushnell Pro X7 Jolt Slope Rangefinder
- SOLE E35 Elliptical
- SOLE E25 Elliptical

Further investigation showed that these products operate with approximately zero profit margin.

While their discount rates remained close to the overall average discount rate (~10%), the absence of sufficient margin caused discounted sales to become unprofitable.

This finding suggests a pricing strategy issue rather than an excessive discounting problem.

Key Business Insights

Financial Insights

- Total revenue exceeded $32.7 million.
- Consumer customers generated more than half of total profits.
- A limited number of categories contributed disproportionately to profitability.
- Puerto Rico was the highest profit-generating state.

Operational Insights

- More than half of all orders experienced delivery delays.
- First Class shipping demonstrated the worst delivery performance despite being a premium service.
- Fraudulent transactions were exclusively associated with bank transfer payments.
- Three products consistently generated losses due to insufficient profit margins.

Strategic Recommendations

- Investigate operational processes affecting First Class shipments.
- Implement additional fraud controls for bank transfer transactions.

- Review pricing strategies for loss-making products.
- Prioritize investment in high-performing customer segments and profitable product categories.
- Monitor delivery performance through operational KPIs to reduce late shipment rates.

Tools & Technologies

- Power BI
- Power Query
- DAX
- Google Sheets
- Google Apps Script
- Snowflake Schema Modeling
- Data Validation & Data Quality Assessment
- Business Intelligence & Supply Chain Analytics